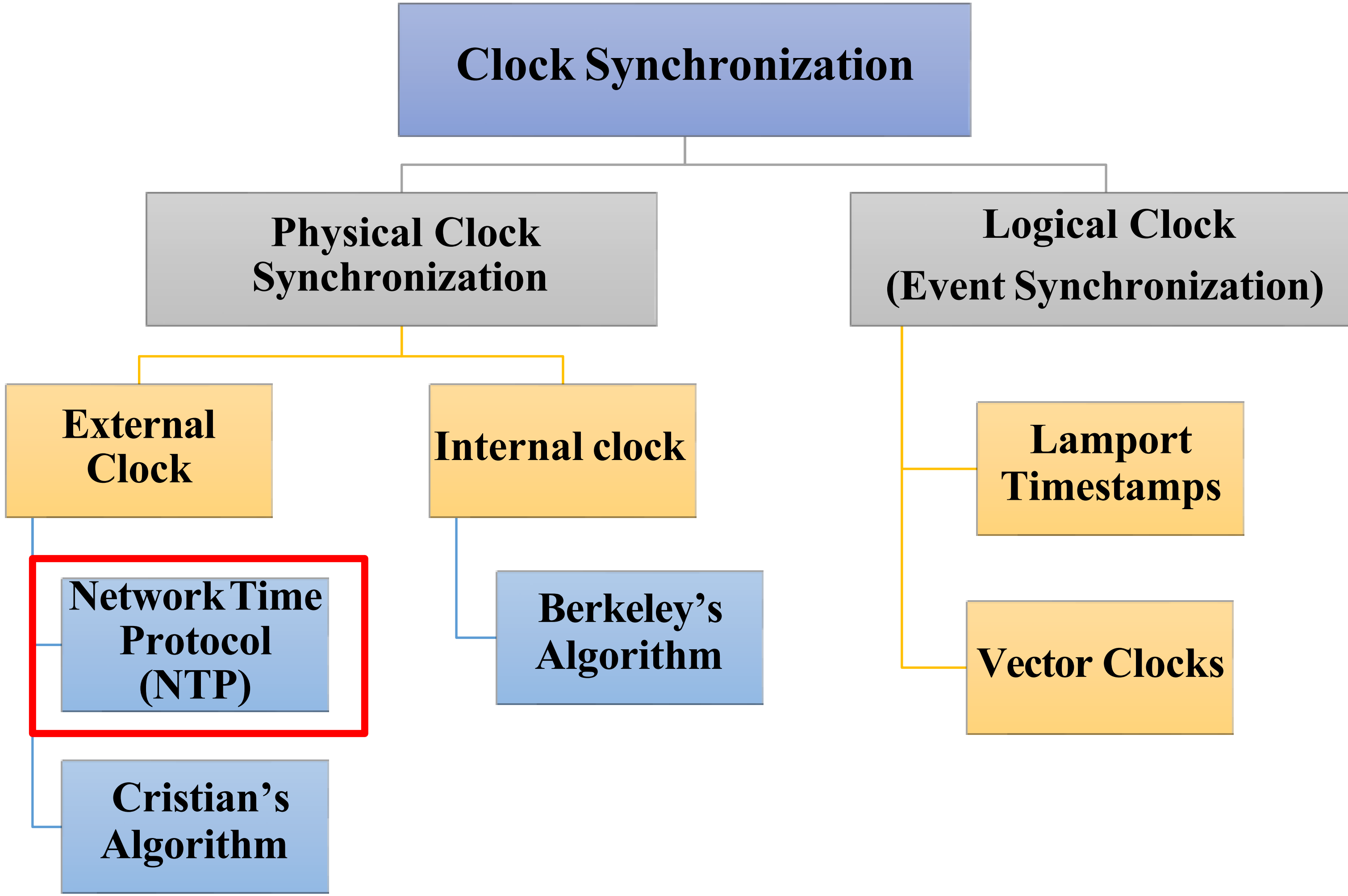
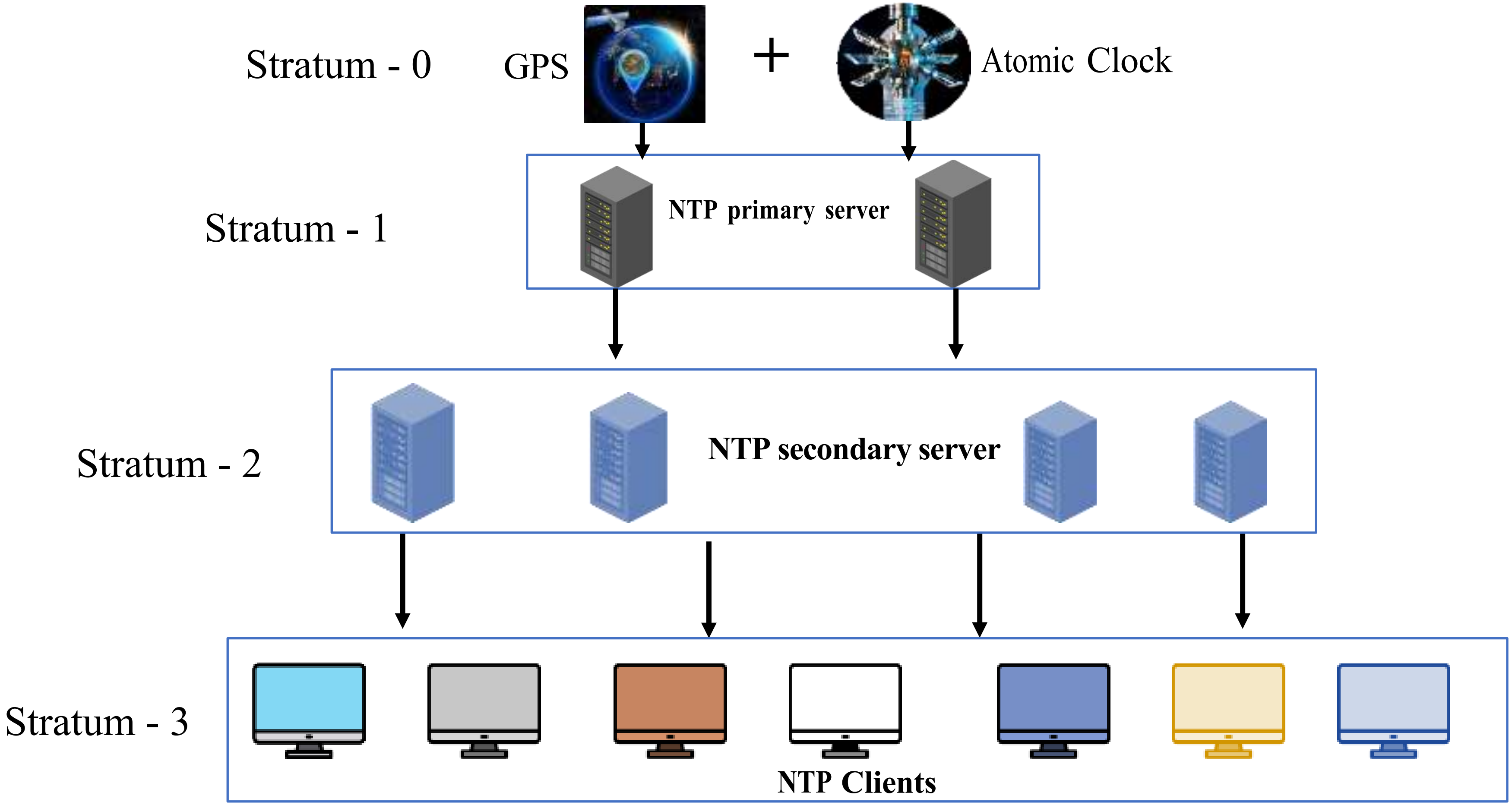
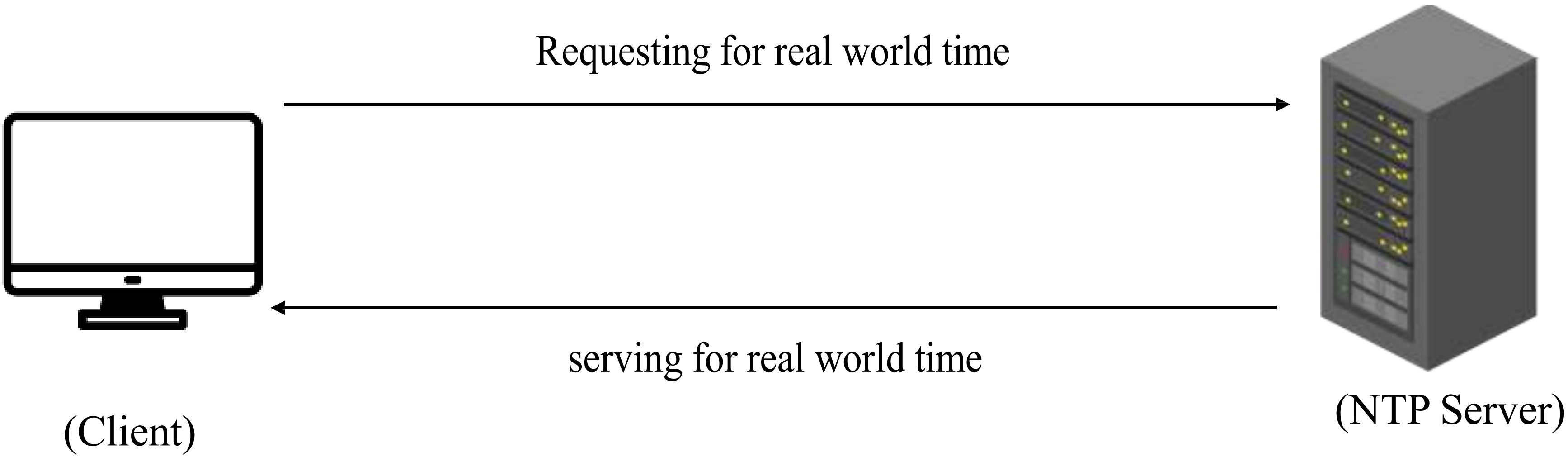


Clock Synchronization
Physical Clock Synchronization
Logical Clock (Event Synchronization)
External Clock
Internal clock
Lamport Timestamps
Network Time Protocol (NTP)
Berkeley's Algorithm
Vector Clocks
Cristian's Algorithm



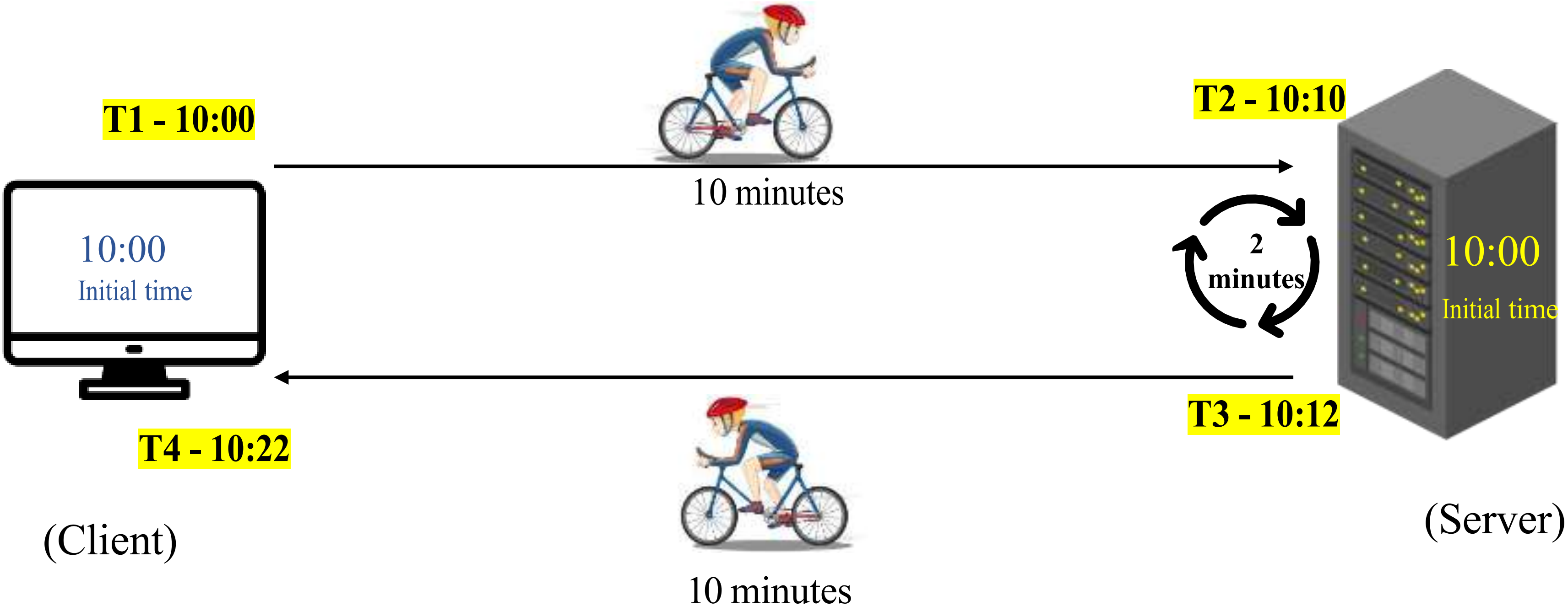
There are 16 possible stratum levels (0–15) used in NTP



We need to calculate:

1. **Round Trip Delay (δ)**:The Round-Trip Delay is the total time taken for a message to travel from the client to the server and back to the client.

2. **Clock Offset (θ):** The Clock Offset is the estimated difference between the client's clock and the server's clock, after compensating for the network transmission delay.



Round Trip Delay (δ) $= (T_4 - T_1) - (T_3 - T_2)$

= (10:22 – 10:00) – (10:12 – 10:10)

=(22-2)

=20 minutes

Clock Offset (θ): $$\frac{(T_2 - T_1) + (T_3 - T_4)}{2}$$

$$= \frac{(10{:}10 - 10{:}00) + (10{:}12 - 10{:}22)}{2}$$

$$= \frac{(10) + (-10)}{2}$$

$$\frac{0}{2} = 0$$

Clock Offset (θ): $$\frac{(T_2 - T_1) + (T_3 - T_4)}{2}$$

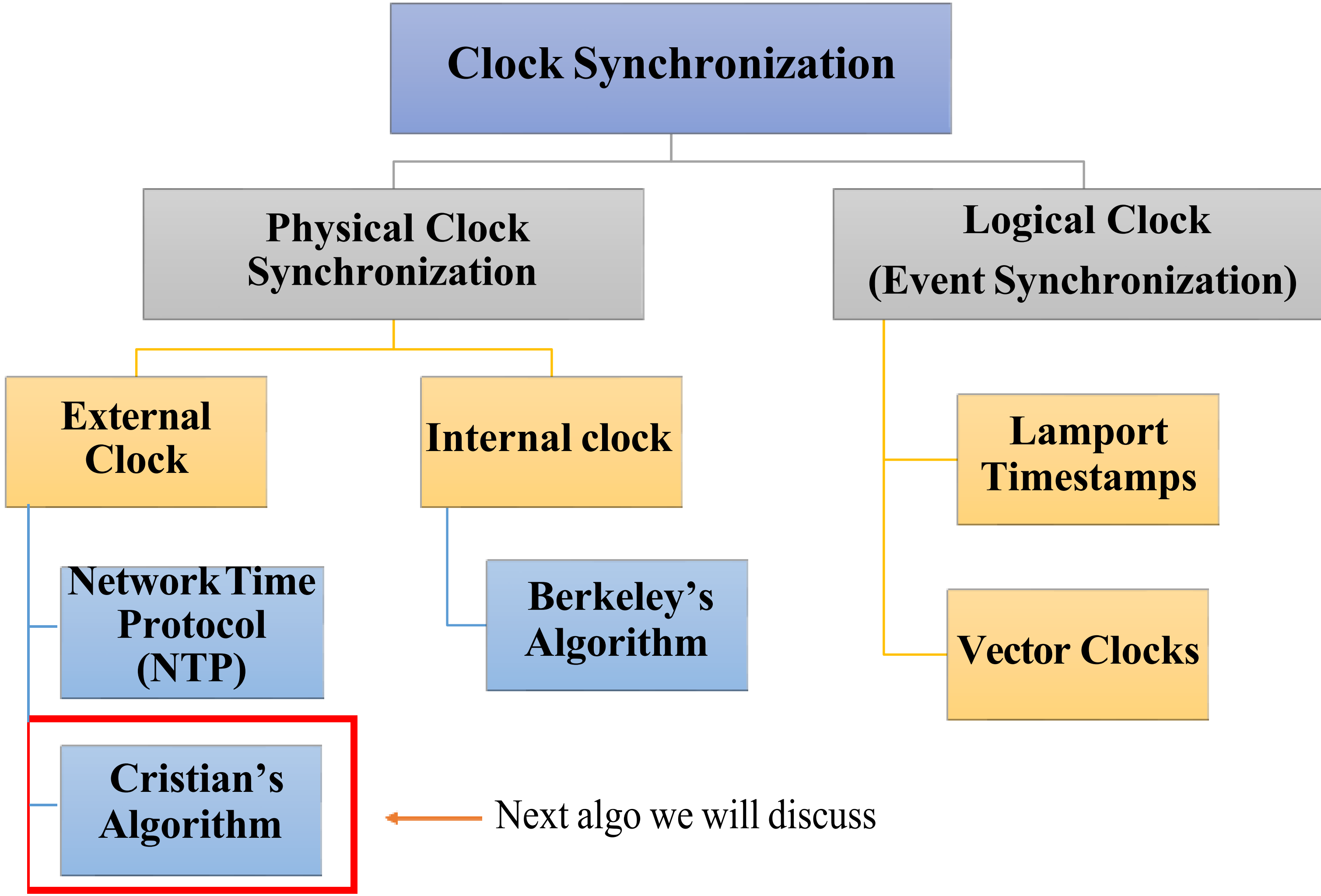
$$= \frac{(10{:}10 - 10{:}10) + (10{:}12 - 10{:}32)}{2}$$

$$= \frac{(0) + (-20)}{2}$$

$$\frac{-20}{2} = -10$$

When time is not synchronized

- If the offset is **positive**, the client's clock is **slow** and needs to move **forward**.
- If the offset is **negative**, the client's clock is **fast** and needs to be **delayed**.
- If the clock offset is **zero**, it means that the client's clock and the server's clock are perfectly **synchronized**.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Next algo we will discuss